Sufficient Conditions on (λ_1, λ_2) for Topology and Separation of Reservoir Attractors

[Supplementary Technical Note]

I BACKGROUND & PROBLEM STATEMENT

Auxilliary Objective 2 asks for conditions on $(\lambda_1, \lambda_2, \rho_{sep})$ under which a minimizer of the combined loss

$$L_{total}(W,U) = \underbrace{\sum_{j=1}^{k-1} L_{field}^{(j)}}_{\text{vector field}} + \underbrace{\lambda_1 \sum_{j=1}^{k-1} L_{topo}^{(j)}}_{\text{topological reg.}} + \underbrace{\lambda_2 L_{sep}(W,U)}_{\text{geometric sep}}$$

simultaneously achieves:

1) Topology: $A_j \simeq S^1$ for each active primitive $j = 1, \dots, k-1$
2) Seperation: $d_H(A_j, A_\ell) > \eta$ for all $j \neq \ell$

The core strategy is an ε-net argument: I show that any (W, U) that violates condition 1 or 2 must have a strictly positive directional derivative of L_{total} pointing towards a better configuration — so it cannot be a local minimizer. Conditions on (λ_1, λ_2) are then the requirement that these gradients are not destroyed by opposing loss terms.

II GEOMETRIC PRELIMINARIES

Persistence and the Topological Gap

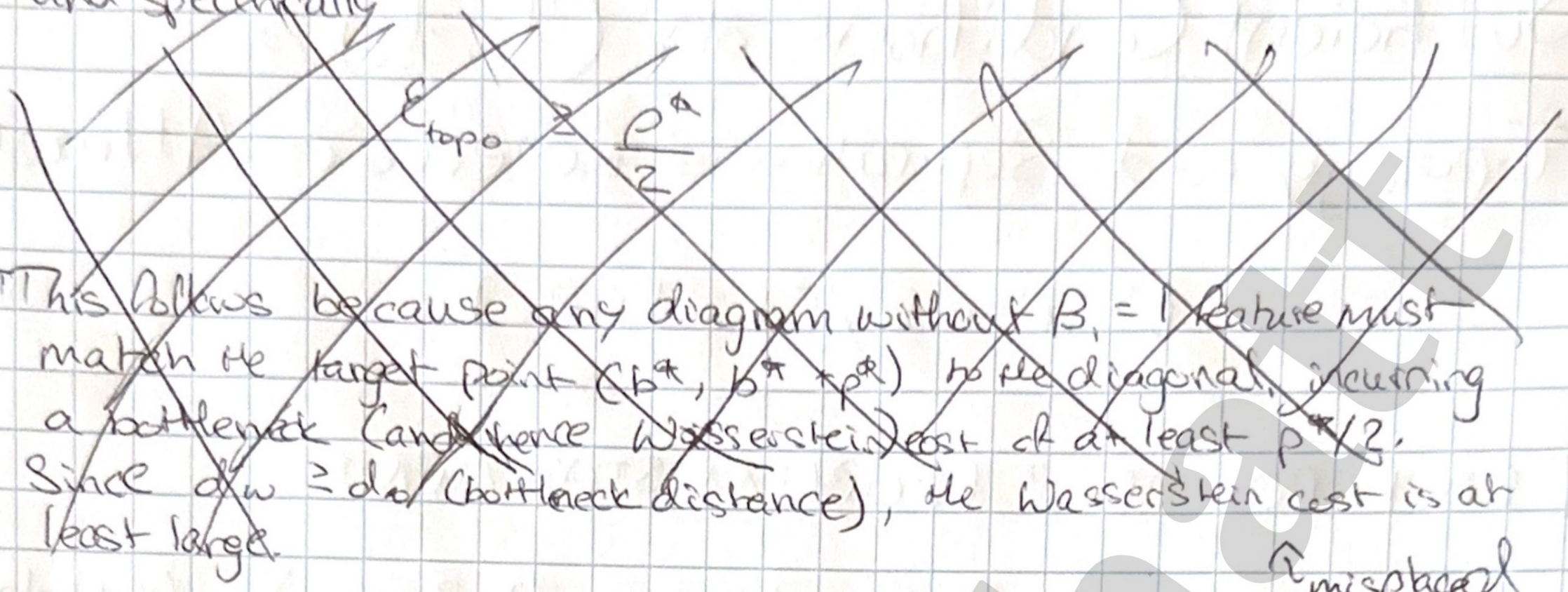
Definition: For reservoir state cloud $\{r^{(j)}(t)\}$, let $PH_1(\{r^{(j)}(t)\}) = \{(b_i, d_i)\}$ be the degree-1 persistence diagram. The dominant persistence is

$$pers_j = \max_i (d_i - b_i)$$

The target diagram consists of a single point at persistence $p^* > 0$.

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and specifically



This follows because any diagram without $\beta_1 = 1$ feature must match the target point $(b^*, b^* + p^*)$ to the diagonal, incurring a bottleneck (and hence Wasserstein) cost of at least $p^*/2$. Since $d_W \geq d_\infty$ (bottleneck distance), the Wasserstein cost is at least large.

$\hat{\tau}_{misplaced}$

(one long-lived S^1) plus the diagonal.

Definition (Topological Gap) Let D_j^{target} consist of a single point at persistence $p^* > 0$ plus the diagonal. The minimum topological defect is

$$\varepsilon_{topo} = \inf\{ d_W(PH_1(\{r\}), D_j^{target}) : \{r\} \text{ has no loop homotopic to } S^1 \} \geq \frac{p^*}{2}$$

Since any diagram without a $\beta_1 = 1$ feature must match the target point to the diagonal, costing at least $p^*/2$ in Wasserstein distance.

Lemma 2.2 (Topology Field Coupling). If $\Lambda_j \neq S^1$, then $L_{field}^{(ij)} \geq \delta_{topo} := c_{dyn} r_j^2$, where r_j is the normal hyperbolicity gap and $c_{dyn} = \det \mathrm{Vol}(\pi_j(\Sigma_j))/N > 0$.

Proof: When Λ_j is a fixed point, the target field rotates tangentially while the reservoir field points inward. The L^2 mismatch over $\pi_j(\Sigma_j)$ is at least r_j, yielding $L_{field}^{(ij)} \geq c_{dyn} r_j^2$. Higher dimensional attractors only increase the mismatch energy. ∎

Lemma 2.3 (Separation-field coupling): If $d_H(\Lambda_j, \Lambda_\ell) \leq \eta$ for some $j \neq \ell$ then

$$L_{field}^{(ij)} + \lambda_1 L_{topo}^{(ij)} + L_{field}^{(\ell)} + \lambda_1 L_{topo}^{(\ell)} \geq \delta_{sep}(\lambda_1) := c_{sep} \rho_{sep}^2 (1 + \lambda_1)$$

where $c_{sep} > 0$ depends on the CCA geometry and spectral gap Δ.

Proof: When attractors overlap within a η-ball, the CCA-structure forces $\| B^T v^{(i)} - B^T v^{(\ell)} \|_2 \geq \rho_{sep} \rho_{min}$, so the reservoir must either

fail to track the target ~~manifold~~ field (contributing to L_{field}) or distort attractor topology (contributing to L_{topo}). The $(1+\lambda_1)$ factor reflects both terms.

I also define bad sets: $B_{topo} = \{(W, U) : \exists j, A_j \neq S^1\}$ and $B_{sep} = \{(W, U) : \exists j \neq l, d_H(A_j, A_l) \leq \eta\}$. On B_{topo}, $L_{topo}^{(j)} \geq \varepsilon_{topo} = \rho^*/2$. On B_{sep}, $L_{sep} \geq \varepsilon_{sep}^2$ where $\varepsilon_{sep} = c_0 \rho_{sep} \eta > 0$ by the CCA separation guarantee.

III MAIN RESULTS

Theorem 3.1 (Sufficient condition on λ_1) Any local minimizer of L_{total} satisfies $A_j \simeq S^1$ for all j if

$$\boxed{\lambda_1 > \frac{\delta_{topo}}{\varepsilon_{topo}} = \frac{2 c_{dm} r_j^2}{\rho^*}}$$

Proof. Suppose $(W^*, U^*) \in B_{topo}$ is a local minimizer. Perturb along the topology correcting direction H (the steepest descent direction for $L_{topo}^{(j)}$, which exists by differentiability of d'_W. The directional derivative satisfies

$$\frac{dL_{total}}{d\varepsilon}\Big|_{\varepsilon=0} \leq \underbrace{\nabla_W L_{field}^{(j)} \cdot H}_{\leq \delta_{topo}/\varepsilon_0} + \underbrace{\lambda_1 \nabla_W L_{topo}^{(j)} \cdot H}_{= -K < 0}$$

where $K > 0$ is the gradient magnitude of d'_W. Normalizing so that $K\varepsilon_0 = \varepsilon_{topo}$, this is negative whenever $\lambda_1 > \delta_{topo}/\varepsilon_{topo}$, contradicting local minimality.

Theorem 3.2 (Sufficient condition on λ_2) Let λ_1 satisfy the previous condition. Then every local minimizer satisfies $d_H(A_j, A_l) > \eta$ for all $j \neq l$ if

$$\boxed{\lambda_2 > \frac{\delta_{sep}(\lambda_1)}{\varepsilon_{sep}^2} = \frac{c_{sep} \rho_{sep}^2 (1+\lambda_1)}{\varepsilon_{sep}^2}}$$

Proof Suppose $(W^*, U^a) \in \mathcal{B}_{sep}$ is a local minimizer. By Theorem 3.1, both $\mathcal{L}_1^*, \mathcal{L}_2^* \simeq S^1$ (topologically correct but geometrically overlapping). Perturb along the canonical separation direction K from the CCA oscillatory subspace:

$$\left.\frac{dL_{total}}{d\varepsilon}\right|_{\varepsilon=0} \leq \delta_{sep}(\lambda_1) - 2\lambda_2 \varepsilon_{sep} \kappa_{sep}.$$

Setting $\kappa_{sep} = \varepsilon_{sep}/2$ (natural scale), this is negative whenever $\lambda_2 > \delta_{sep}(\lambda_1)/\varepsilon_{sep}^2$, giving the contradiction. ∎

IV JOINT FEASIBILITY REGION

Corollary 4.1 (Joint feasibility region) Both conditions hold at every local minimizer provided $(\lambda_1, \lambda_2) \in \mathcal{F}$, the wedge

$$\mathcal{F} = \left\{ (\lambda_1, \lambda_2) : \lambda_1 > \lambda_1^*, \text{ and } \lambda_2 > \frac{c_{sep}}{c_0^2 \eta^2} \lambda_1 \right\}, \quad \lambda_1^* = \frac{2\varepsilon_{dyn} T^2}{\rho^a}$$

using $\varepsilon_{sep} \geq c_0 \rho_{sep} \eta$ for $\lambda_1 \gg \lambda_1^*$. The linearity $\lambda_2 \propto \lambda_1$ reflects that stronger topological regularization increases the cost of separation modes, requiring a proportionally larger separation penalty

Proposition 4.2 (Role of ρ_{sep}):

1) λ_1^* is independent of ρ_{sep}; it depends only on T and ρ^a.
2) $\lambda_2^*(\lambda_1) = c_{sep}\lambda_1/(c_0^2\eta^2)$ is also independent of ρ_{sep} to leading order (the ρ_{sep}^2 factors cancel)
3) The upper bound on λ_2 (field-matching dominance) scales as ρ_{sep}^2: larger ρ_{sep} allows a wider feasible range.

For $\mathcal{F}$ to be nonempty with the upper bound, we need

$$\rho_{sep} > \frac{1}{c_0 \eta} \sqrt{\frac{2 c_{sep} c_{dyn} T^2}{\rho^a}}$$